BIO 005 Human Physiology · Week 7 notes · Dr. Sharilyn Rennie

Twitches, summation and motor units

The written notes for the twitches, summation and motor units walkthrough. Week 7 competencies 6 to 10, with part of 16.

To print: these notes as a two-column PDF. Already printed the Week 7 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the walkthrough.

Step by step: from one shock to a full contraction

In your textbook: Silverthorn Chapter 13, the parts on the muscle twitch, summation and tetanus, and motor units.

Dr. Rennie's videos for this part: What changes the force a muscle produces.

One twitch, step by step

1. A single stimulus, from the motor neuron or from a stimulator, fires one action potential in the muscle fiber. It lasts only a few milliseconds.
2. Latent period: the action potential spreads down the T tubules, the Ca2+ release channels of the sarcoplasmic reticulum open, Ca2+ binds troponin, tropomyosin moves aside, and crossbridges begin to form. Tension has not risen yet.
3. Contraction phase: Ca2+ is high, more and more crossbridges cycle, and tension rises to a peak.
4. The Ca2+ pump in the sarcoplasmic reticulum membrane, SERCA, uses ATP to pull Ca2+ back out of the cytosol. Ca2+ starts to fall before tension peaks.
5. Relaxation phase: as Ca2+ leaves troponin, tropomyosin covers the binding sites again, crossbridges let go, and tension falls back to zero. This is the longest phase.

Summation and tetanus, step by step

1. A second action potential arrives before the fiber has fully relaxed.
2. Some Ca2+ from the first one is still in the cytosol, because SERCA has not finished pumping it back.
3. The second release adds to what is left over, so Ca2+ rises higher than after one action potential.
4. More troponin has Ca2+ bound, more crossbridges pull at once, and tension climbs above a single twitch: summation.
5. At a higher frequency, the pumps fall further behind. Tension climbs and dips with each stimulus: unfused tetanus.
6. At the highest frequency, Ca2+ stays high between stimuli and tension reaches a smooth plateau: fused tetanus, the strongest contraction the fiber can make.
7. The action potentials themselves never add together. They are all-or-none and have a refractory period.

Recruiting motor units, step by step

1. The brain or a reflex sends excitatory input to the pool of motor neurons for one muscle in the ventral horn of the spinal cord.
2. The smallest motor neurons reach threshold first, because they need the least synaptic input. They fire their small motor units of slow oxidative fibers. The force is small and finely controlled.
3. As the input grows, medium motor neurons reach threshold and add their fast oxidative-glycolytic fibers.
4. With still more input, the largest motor neurons fire and add their many fast glycolytic fibers. The force jumps.
5. Motor neurons that are already active also fire faster, so their twitches summate toward tetanus.
6. In a long steady contraction, the nervous system takes turns among motor units so some rest while others work.

Lifting a load, step by step

1. The muscle is stimulated and tension starts to rise. At first the muscle does not change length.
2. If the load is smaller than the most tension the muscle can make, tension rises until it equals the load.
3. Then the muscle shortens and moves the load, with tension steady at the level of the load: a concentric isotonic contraction.
4. If the load is more than the muscle can make, tension rises to its maximum and the muscle stays the same length: an isometric contraction.
5. If the load is a little more than the tension the muscle is making, the active muscle is stretched while it keeps pulling: an eccentric contraction.

The muscle twitch

In your textbook: Silverthorn Chapter 13, the part on the muscle twitch.

Dr. Rennie's videos for this part: The muscle twitch and how tension is measured, The phases of a myogram.

A twitch is a single contraction and relaxation cycle of a muscle fiber, produced by a single action potential. A twitch tracing plots tension against time, with the stimulus marked on the time axis.

Phase	What is happening	What limits how long it lasts
Latent period	Tension is flat. The action potential spreads into the T tubules, Ca2+ is released from the sarcoplasmic reticulum, binds troponin, and tropomyosin moves aside	Excitation-contraction coupling, the time it takes to get Ca2+ onto troponin and crossbridges cycling
Contraction phase	Tension rises to a peak as more crossbridges cycle	How long Ca2+ stays high, and how fast the crossbridges cycle
Relaxation phase	Tension falls back to zero	How fast SERCA pumps Ca2+ back into the sarcoplasmic reticulum. It is the longest phase

- The muscle action potential lasts only a few milliseconds and is finished before tension even starts to rise.
- Tension follows the Ca2+ in the cytosol, not the action potential. Ca2+ starts falling before tension peaks.
- If SERCA is slowed, Ca2+ stays high longer, the relaxation phase gets much longer and the peak a little higher. The latent period does not change.
- Relaxation costs energy: SERCA uses ATP to move Ca2+ against its gradient.
- Because the electrical event is so short and the contraction so long, a second action potential can arrive while the fiber is still contracting. That makes summation possible.

Twitch

One contraction and relaxation of a fiber in response to one action potential.

Latent period

The delay between the muscle action potential and the start of tension, while excitation-contraction coupling happens.

SERCA

The Ca2+ ATPase in the sarcoplasmic reticulum membrane that pumps Ca2+ out of the cytosol and ends a contraction.

Summation and tetanus

In your textbook: Silverthorn Chapter 13, the part on summation and tetanus.

Dr. Rennie's videos for this part: How a muscle responds to faster stimuli: summation and tetanus.

The force of one fiber is graded by how often it is stimulated. The faster the stimuli come, the less time SERCA has to clear Ca2+ between them.

Stimulus frequency	Ca2+ between stimuli	Tension
Low	Falls all the way back to rest	Separate twitches
A second stimulus during relaxation	Some is left over, so the second release adds to it	Summation: the second contraction climbs higher than one twitch
Higher	Never falls to rest, rises and falls with each stimulus	Unfused (incomplete) tetanus: tension climbs and dips with each stimulus
Highest	Stays high	Fused (complete) tetanus: a smooth plateau, the most tension the fiber can make

- Summation comes from Ca2+ building up in the cytosol and more crossbridges pulling at once. It does not come from action potentials adding together; they are all-or-none and have a refractory period.
- In fused tetanus, the tension stops dipping before the Ca2+ does. Ca2+ still rises and falls a little with each stimulus, but never drops low enough for many crossbridges to let go.
- Everyday movements use trains of action potentials and tetanic contractions, not single twitches.
- The disease tetanus is something else: a toxin from Clostridium tetani blocks inhibitory neurons in the spinal cord, so motor neurons fire without control and muscles lock in spasm.

Summation

A contraction added onto one that has not finished relaxing, giving more tension than a single twitch.

Unfused tetanus

Tension that stays raised but rises and falls with each stimulus. Also called incomplete tetanus.

Fused tetanus

A smooth, maximal plateau of tension at a high stimulus frequency. Also called complete tetanus.

Motor units and recruitment

In your textbook: Silverthorn Chapter 13, the part on motor units.

Dr. Rennie's videos for this part: Motor units, load and resistance, Motor unit recruitment.

A motor unit is one somatic motor neuron and all the skeletal muscle fibers it innervates. Each fiber is innervated by only one motor neuron, but one motor neuron branches to many fibers.

- When the motor neuron fires, every fiber in its unit gives a full twitch, and no other fibers do.
- The fibers of one unit are spread through the muscle, not bunched together, so even one small unit pulls on the whole muscle evenly.
- All the fibers in one motor unit are the same fiber type.
- Muscles for fine, precise movement, such as the eye muscles and the small muscles of the hand, have small motor units with only a few fibers each. Each added unit adds a small step of force.
- Muscles for gross movement and support, such as the calf muscle, have large motor units with hundreds to thousands of fibers each. Each added unit adds a big step of force.

Motor neuron	Its fibers	Recruited
Small	A few slow oxidative fibers: slow, fatigue-resistant, small force	First, for light efforts
Medium	More fast oxidative-glycolytic fibers: fast, fairly fatigue-resistant	As effort increases
Large	Many fast glycolytic fibers: fast, the most force, tire quickly	Last, for the heaviest efforts

Recruitment

Adding more motor units to increase the force of a whole muscle.

Size principle

Motor units are recruited in order of size, smallest first, because small motor neurons reach threshold with less synaptic input.

Asynchronous recruitment

Taking turns among motor units during a long steady contraction, so some rest while others work and the muscle is slower to fatigue.

The nervous system grades the force of a whole muscle in two ways: how many motor units are active (recruitment) and how fast each one fires (summation and tetanus). Picking up a paper cup uses only small units; lifting a gallon of milk adds medium and then large units.

Stimulus voltage in the lab (Competency 16)

- In PhysioEx Exercise 2 a stimulator shocks a whole muscle, and you raise the voltage step by step.
- Below threshold, no fibers respond and there is no twitch.
- Threshold stimulus: the lowest voltage that produces a visible twitch, because only a few fibers reach threshold.
- Between threshold and maximal, more fibers respond with each step, so the twitch gets stronger. Each fiber still follows the all-or-none rule.
- Maximal stimulus: the lowest voltage at which every fiber responds. Higher voltages add no more force.
- Raising the voltage on a whole muscle works much like recruiting motor units in the body: force grows because more fibers take part.

Length and tension

In your textbook: Silverthorn Chapter 13, the part on the length-tension relationship.

Dr. Rennie's videos for this part: The length-tension relationship.

The active tension a fiber makes depends on the length of its sarcomeres before it contracts, because crossbridges form only where myosin heads overlap thin filaments.

Sarcomere length	Filaments	Active tension
Too short (about 1.5 micrometers)	Thin filaments from the two sides overlap each other and get in each other's way; thick filaments press against the Z discs	Lower
Best (about 2.0 to 2.2 micrometers)	Every myosin head can reach a thin filament, and the thin filaments have room to slide	Greatest
Too long (about 3.2 micrometers)	Thin filaments barely overlap the ends of the thick filaments, so few heads can reach actin	Lower, and zero once there is no overlap

- Tension falls at both ends of the curve, for different reasons: too short, the filaments jam; too long, there is too little overlap.
- The 2.0 to 2.2 micrometer plateau in the textbook curve comes from classic measurements of frog muscle. Human thin filaments are a little longer, so human muscle works best at a slightly longer sarcomere length.
- In the body, the way muscles attach to bones keeps their resting length close to the best overlap.
- Bend your wrist all the way forward and make a fist: the finger flexors are already short, and the grip is weaker.
- A muscle pulled far past its resting length can form few crossbridges and makes only a little force.

Length–tension relationship

The active tension a sarcomere makes depends on its length when it is stimulated, and is greatest at the length with the best filament overlap.

Isometric and isotonic contraction

In your textbook: Silverthorn Chapter 13, the part on the mechanics of body movement, isotonic and isometric contractions.

Dr. Rennie's videos for this part: Isometric and isotonic contraction, Muscle tension and its components, Isotonic and isometric contractions: concentric and eccentric, The force-velocity relationship, Muscle load and tension.

Contraction	Muscle length	Tension compared with the load	Example
Isotonic, concentric	Shortens	Equal to the load, which is less than the muscle's maximum	Curling a dumbbell up
Isometric	Stays the same	Equal to the load (holding still), or the load is more than the muscle can make	Holding a weight still; pushing on a wall
Eccentric	Lengthens while active	A little less than the load	Lowering a dumbbell slowly; walking downstairs

- In every contraction that moves a load, tension first rises until it equals the load, with no change in length, and only then does the muscle shorten.
- An isometric contraction uses ATP the whole time, even though nothing moves.
- In an isometric contraction the sarcomeres shorten slightly as they stretch the tendons and connective tissue, the series elastic elements, but the muscle as a whole does not change length.

- Eccentric work, such as walking downhill or lowering weights, is the kind most likely to cause delayed onset muscle soreness a day or two later.

The force-velocity relationship

- With the muscle fully activated, the speed of shortening falls as the load rises.
- With almost no load, the muscle shortens fastest.
- When the load equals the most tension the muscle can make, the speed is zero and the contraction is isometric.
- This is why you can throw a light ball much faster than a heavy one.

Isotonic contraction

A contraction that moves a load. Concentric when the muscle shortens, eccentric when the active muscle lengthens.

Isometric contraction

A contraction that makes tension without changing the length of the muscle.

Force–velocity relationship

The heavier the load, the slower a muscle shortens; at the muscle's maximum tension, shortening stops.